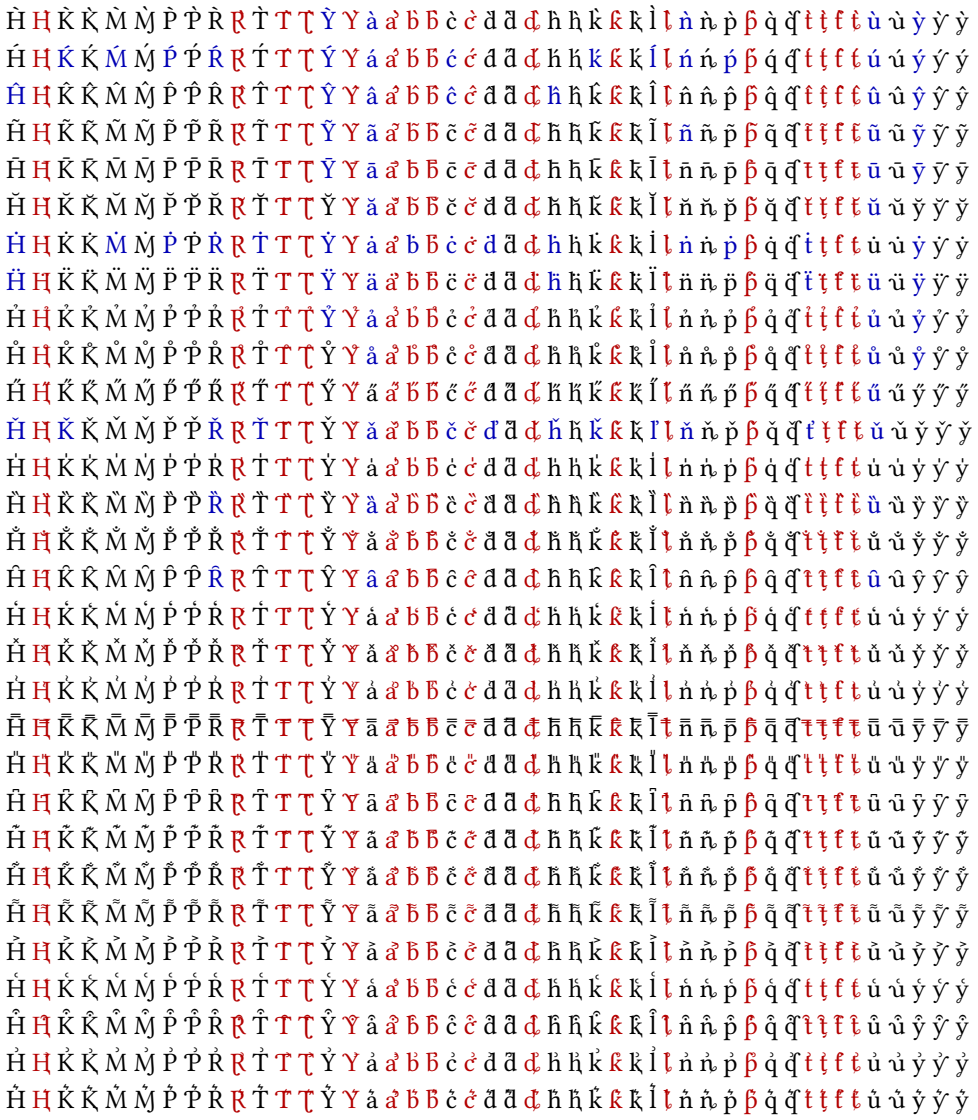
Group 4 derived letters: test sheet

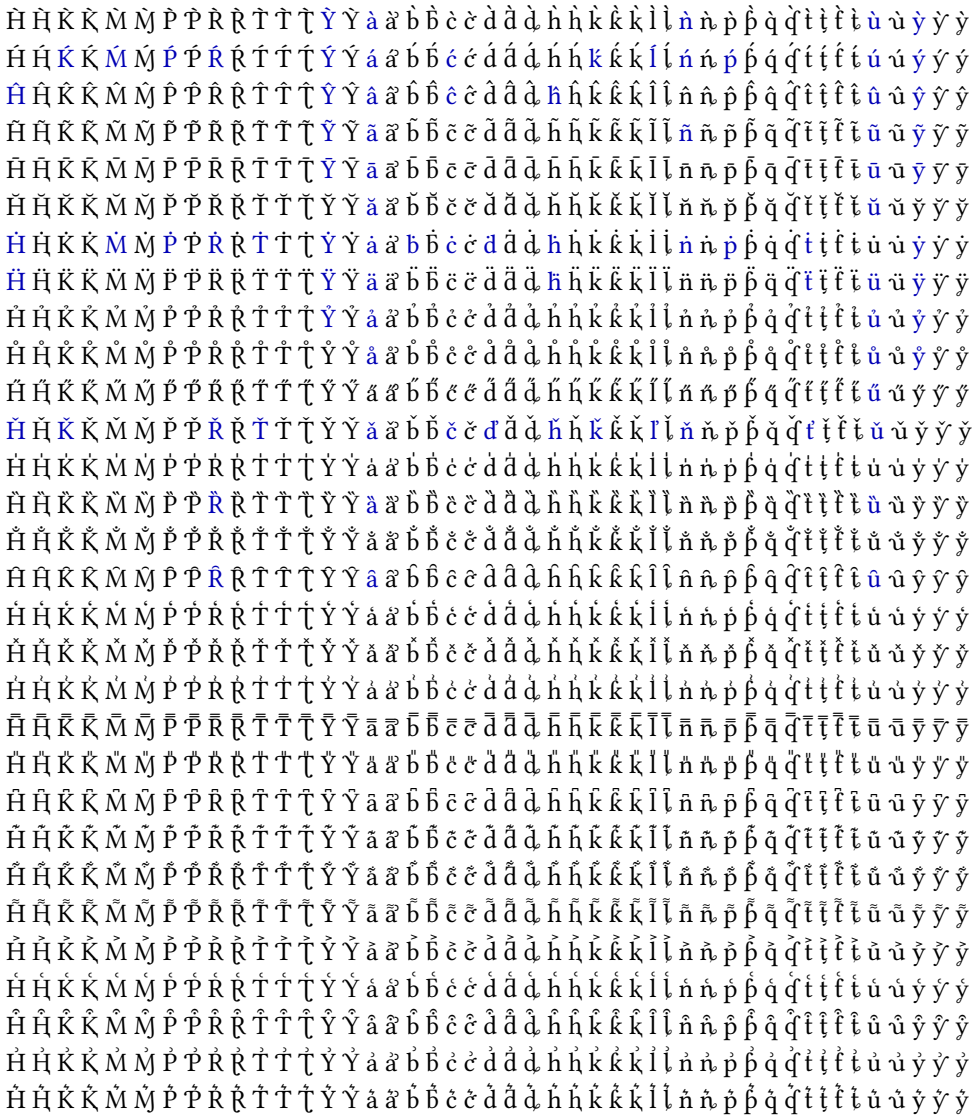
Letters with a hook, tail, curl, topbar or descender, added to stage 8 on 2026-11-11 (`data/derived_letters.py`), each after its base letter. Each grid row is one IPA mark on every letter; black is positioned by an anchor, red is the fallback (no anchor), blue a precomposed glyph, gray a missing glyph. The original fonts come first, then the patched fonts.

letters	rule	why
Ƥ ƀ đ Ƴ ƴ ƙ ƈ ƈ̉ ʊ	inherit both (fit)	the hook, topbar or ring stays clear of the marks; italic and semibold italic redrawings keep stage 6's candidate
T f	stem shift (`stem_axis`)	the hook at the end of T's bar spoils the fit above; f's hook replaces the top of t's stem
þ ʠ f above	`derived_y`: X from p, q, t, Y at the ascender reference	the hook rises from the stem to the ascender, under the mark (it would move Y by $+102$ to $+256$)
ţ Ƭ ḍ ɬ ɳ ŧ Ɽ Ḥ ḥ Ḳ ḳ Ɱ ɣ	`bottom_tail`	stems matched above the tail; below, the candidate's Y where the tail is under the mark, else the base letter's (Ḥ)
Ƶ ʒ	not derived	Z has no stems: stage 6's candidate

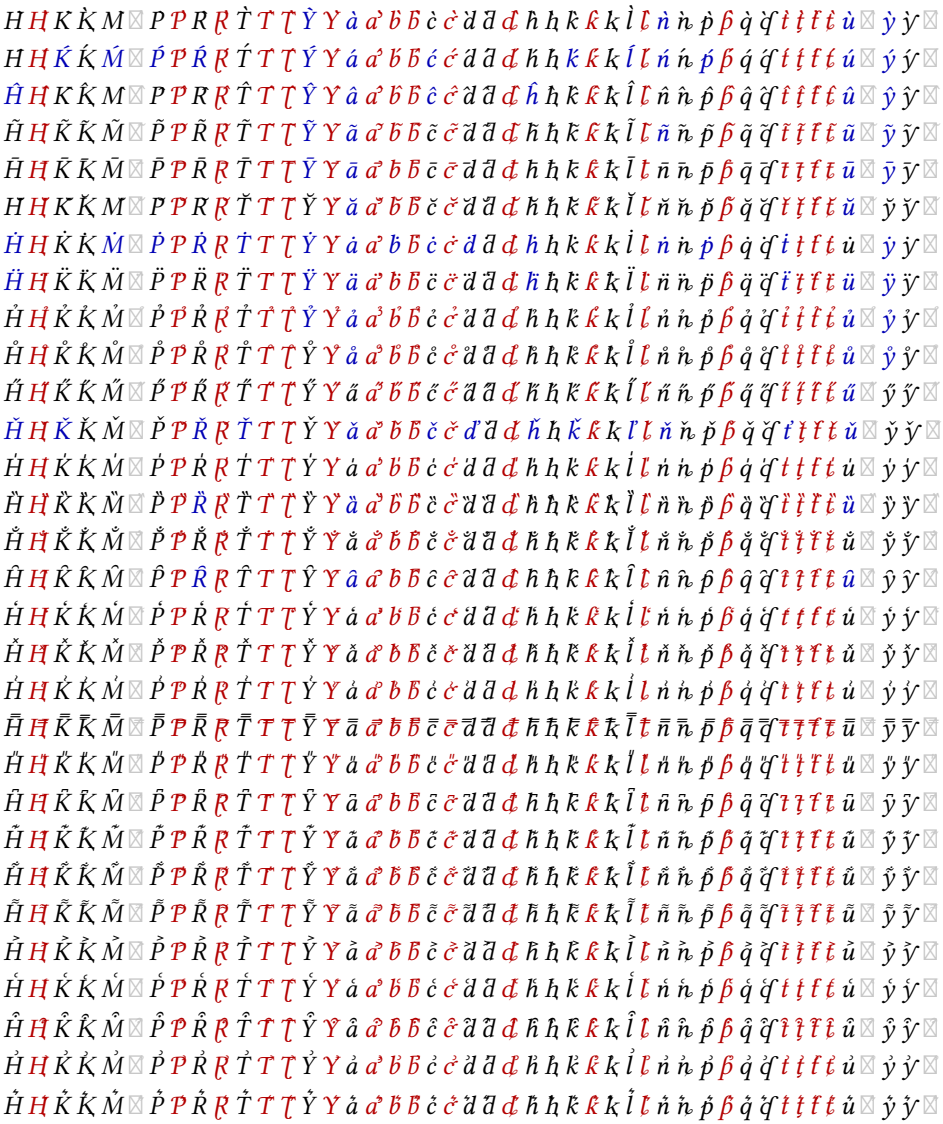
Regular, Group 4, above, Above



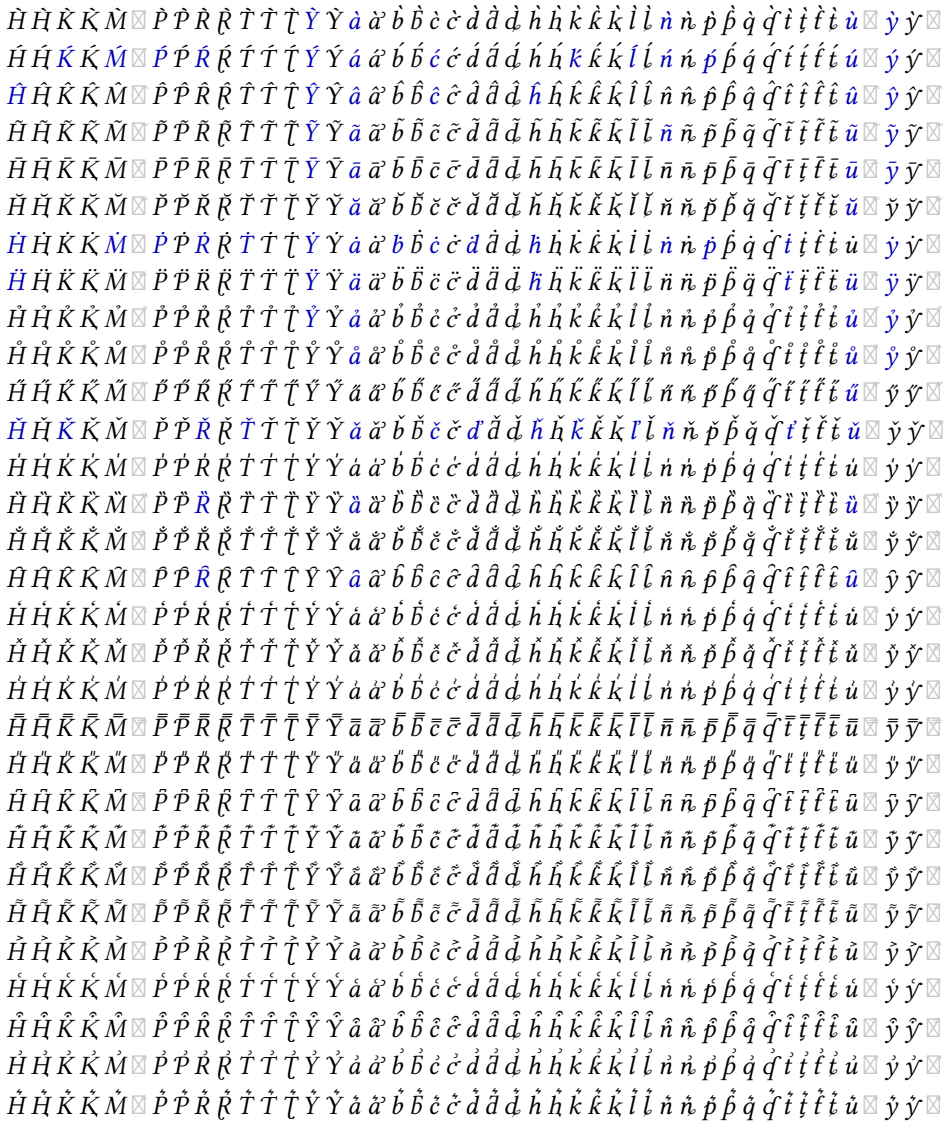
Regular patched, Group 4, above, Above



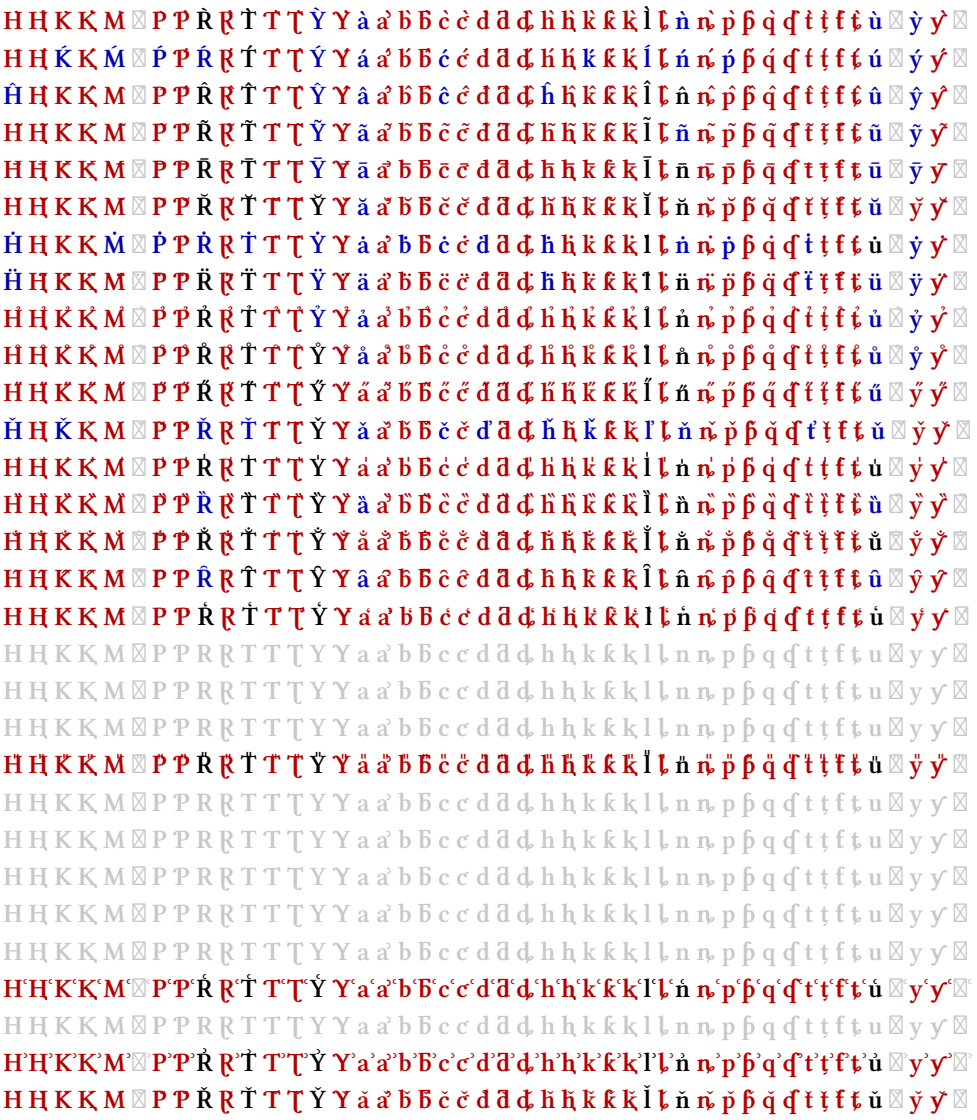
Italic, Group 4, above, Above



Italic patched, Group 4, above, Above



Semibold, Group 4, above, Above



Semibold patched, Group 4, above, Above

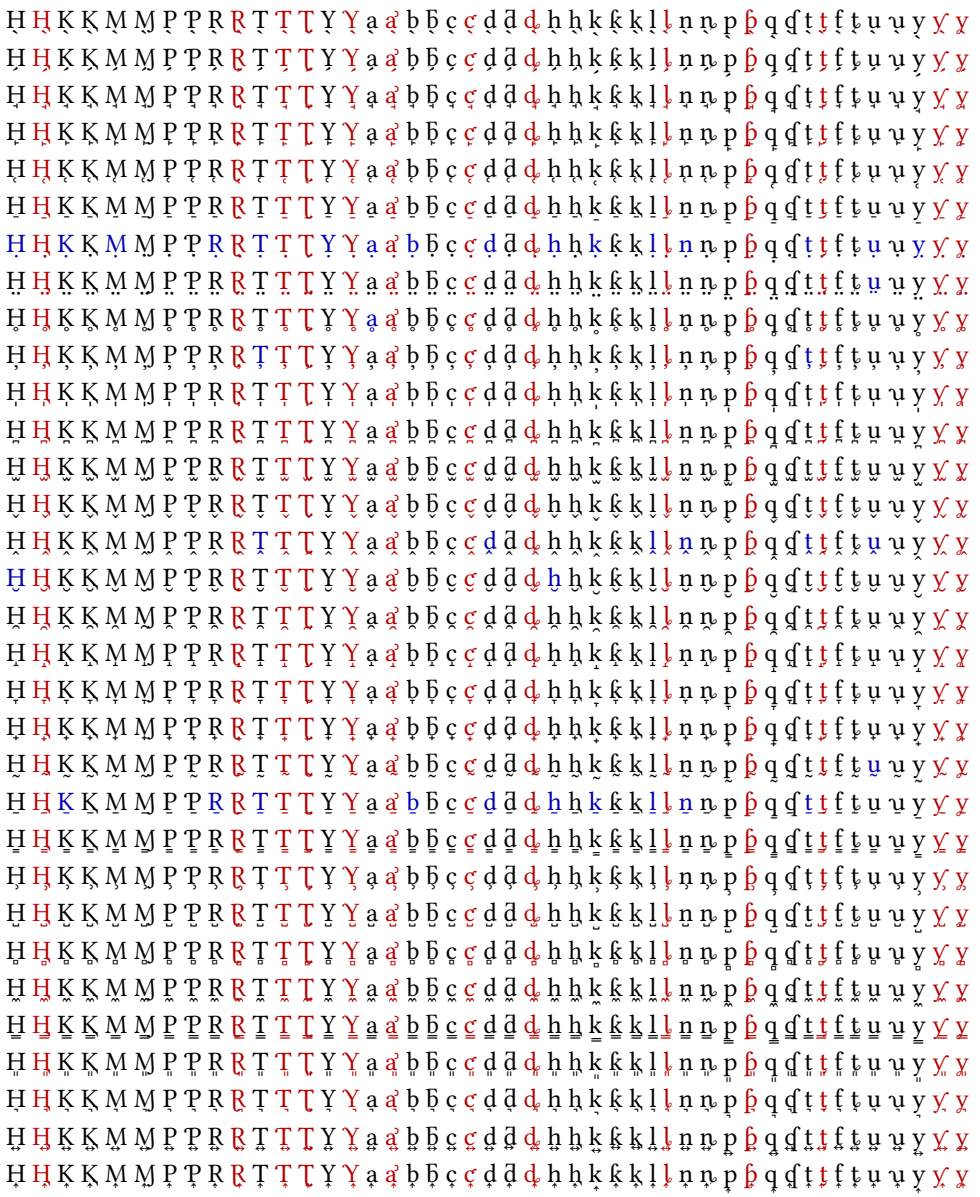
Ȟ Ḥ Ḵ Ḳ Ṁ Ṕ Ṗ Ṛ Ṝ Ṯ Ṯ Ṱ Ẏ Ỳ à ạ̀ b̀ b̥̀ c̀ c̥̀ d̀ d̥̀ ḥ̀ ḥ̀ k̀ k̥̀ l̀ l̥̀ ǹ ǹ̥ p̀ p̥̀ q̀ q̥̀ t̀ t̥̀ t̥̀ ù ỳ ỷ
Ḥ Ḥ́ Ḵ Ḳ Ḿ Ṕ Ṗ Ṛ Ṝ Ṯ Ṯ Ṱ Ý Ỹ á ạ́ b́ b̥́ ć ć̥ d́ d̥́ ḥ́ ḥ́ ḱ ḱ̥ ĺ ĺ̥ ń ń̥ ṕ ṕ̥ q́ q̥́ t́ t̥́ t́ ú ý ỷ
Ĥ Ḥ̂ Ḵ Ḳ̂ Ṁ Ṕ Ṗ Ṛ Ṝ Ṯ Ṯ Ṱ Ŷ Ỹ â ậ b̂ b̥̂ ĉ ĉ̥ d̂ d̥̂ ĥ ḥ̂ k̂ k̥̂ l̂ l̥̂ n̂ n̥̂ p̂ p̥̂ q̂ q̥̂ t̂ t̥̂ t̂ û ŷ ỷ
Ȟ Ḥ̌ Ḵ Ḳ̌ Ṁ Ṕ Ṗ Ṛ Ṝ Ṯ Ṯ Ṱ Ỹ Ỹ ã ạ̃ b̃ b̥̃ c̃ c̥̃ d̃ d̥̃ h̃ ḥ̃ k̃ k̥̃ l̃ l̥̃ ñ ñ̥ p̃ p̥̃ q̃ q̥̃ t̃ t̥̃ t̃ ũ ỹ ỷ
Ħ Ḥ̄ Ḵ Ḳ̄ Ṁ Ṕ Ṗ Ṛ Ṝ Ṯ Ṯ Ṱ Ȳ Ỹ ā ạ̄ b̄ b̥̄ c̄ c̥̄ d̄ d̥̄ h̄ ḥ̄ k̄ k̥̄ l̄ l̥̄ n̄ n̥̄ p̄ p̥̄ q̄ q̥̄ t̄ t̥̄ t̄ ū ȳ ỷ
Ȟ Ḥ̆ Ḵ Ḳ̆ Ṁ Ṕ Ṗ Ṛ Ṝ Ṯ Ṯ Ṱ Ў Ỹ ă ặ b̆ b̥̆ c̆ c̥̆ d̆ d̥̆ h̆ ḥ̆ k̆ k̥̆ l̆ l̥̆ n̆ n̥̆ p̆ p̥̆ q̆ q̥̆ t̆ t̥̆ t̆ ŭ ў ỷ
Ḣ Ḥ̇ Ḵ̇ Ḳ̇ Ṁ Ṗ Ṗ Ṛ Ṝ Ṫ Ṫ Ṱ Ẏ Ỹ ȧ ạ̇ ḃ ḅ ċ ċ̥ ḋ ḍ ḣ ḥ̇ k̇ k̥̇ l̇ l̥̇ ṅ ṅ̥ ṗ ṗ̥ q̇ q̥̇ ṫ ṭ̇ ṫ u̇ ẏ ỷ
Ḧ Ḥ̈ Ḵ̈ Ḳ̈ Ṁ Ṗ Ṗ Ṛ Ṝ Ṯ Ṯ Ṱ Ÿ Ỹ ä ạ̈ b̈ b̥̈ c̈ c̥̈ d̈ d̥̈ ẖ ḥ̈ k̈ k̥̈ l̈ l̥̈ n̈ n̥̈ p̈ p̥̈ q̈ q̥̈ ẗ ṭ̈ ẗ ü ÿ ỷ
Ȟ Ḥ̊ Ḵ̊ Ḳ̊ Ṁ Ṕ Ṗ Ṛ Ṝ Ṯ Ṯ Ṱ Ẏ Ỹ å ạ̊ b̊ b̥̊ c̊ c̥̊ d̊ d̥̊ h̊ ḥ̊ k̊ k̥̊ l̊ l̥̊ n̊ n̥̊ p̊ p̥̊ q̊ q̥̊ t̊ ṭ̊ t̊ ů ẏ ỷ
Ȟ Ḥ̊ Ḵ̊ Ḳ̊ Ṁ Ṕ Ṗ Ṛ Ṝ Ṯ Ṯ Ṱ Ẏ Ỹ å ạ̊ b̊ b̥̊ c̊ c̥̊ d̊ d̥̊ h̊ ḥ̊ k̊ k̥̊ l̊ l̥̊ n̊ n̥̊ p̊ p̥̊ q̊ q̥̊ t̊ ṭ̊ t̊ ů ẏ ỷ
Ȟ Ḥ̋ Ḵ̋ Ḳ̋ Ṁ Ṕ Ṗ Ṛ Ṝ Ṯ Ṯ Ṱ Ÿ Ỹ ắ ạ̋ b̋ b̥̋ c̋ c̥̋ d̋ d̥̋ h̋ ḥ̋ k̋ k̥̋ l̋ l̥̋ n̋ n̥̋ p̋ p̥̋ q̋ q̥̋ t̋ ṭ̋ t̋ ű ÿ ỷ
Ȟ Ḥ̌ Ḵ̌ Ḳ̌ Ṁ Ṕ Ṗ Ṛ Ṝ Ṯ Ṯ Ṱ Ў Ỹ ă ặ b̆ b̥̆ c̆ c̥̆ ď d̥̆ h̆ ḥ̆ k̆ k̥̆ l̆ l̥̆ ň n̥̆ p̆ p̥̆ q̆ q̥̆ t̆ ṭ̆ t̆ ŭ ў ỷ
Ḣ Ḥ̇ Ḵ̇ Ḳ̇ Ṁ Ṗ Ṗ Ṛ Ṝ Ṫ Ṫ Ṱ Ẏ Ỹ ȧ ạ̇ ḃ ḅ ċ ċ̥ ḋ ḍ ḣ ḥ̇ k̇ k̥̇ l̇ l̥̇ ṅ ṅ̥ ṗ ṗ̥ q̇ q̥̇ ṫ ṭ̇ ṫ u̇ ẏ ỷ
Ȟ Ḥ̃ Ḵ̃ Ḳ̃ Ṁ Ṕ Ṗ Ṛ̃ Ṝ Ṯ̃ Ṯ Ṱ Ỹ Ỹ ã ạ̃ b̃ b̥̃ c̃ c̥̃ d̃ d̥̃ h̃ ḥ̃ k̃ k̥̃ l̃ l̥̃ ñ ñ̥ p̃ p̥̃ q̃ q̥̃ t̃ ṭ̃ t̃ ũ ỹ ỷ
Ȟ Ḥ̌ Ḵ̌ Ḳ̌ Ṁ Ṕ Ṗ Ṛ̂ Ṝ Ṯ Ṯ Ṱ Ỹ Ỹ â ậ b̂ b̥̂ ĉ ĉ̥ d̂ d̥̂ ĥ ḥ̂ k̂ k̥̂ l̂ l̥̂ n̂ n̥̂ p̂ p̥̂ q̂ q̥̂ t̂ ṭ̂ t̂ û ŷ ỷ
Ḥ́ Ḥ́ Ḵ́ Ḳ́ Ṁ Ṕ Ṗ Ṛ Ṝ Ṯ Ṯ Ṱ Ý Ỹ á ạ́ b́ b̥́ ć ć̥ d́ d̥́ ḥ́ ḥ́ ḱ ḱ̥ ĺ ĺ̥ ń ń̥ ṕ ṕ̥ q́ q̥́ t́ t̥́ t́ ú ý ỷ
Ȟ Ḥ̌ Ḵ̌ Ḳ̌ Ṁ Ṕ Ṗ Ṛ Ṝ Ṯ Ṯ Ṱ Ў Ỹ ă ặ b̆ b̥̆ c̆ c̥̆ d̆ d̥̆ h̆ ḥ̆ k̆ k̥̆ l̆ l̥̆ n̆ n̥̆ p̆ p̥̆ q̆ q̥̆ t̆ ṭ̆ t̆ ŭ ў ỷ
Ḥ̇ Ḥ̇ Ḵ̇ Ḳ̇ Ṁ Ṗ Ṗ Ṛ Ṝ Ṫ Ṫ Ṱ Ẏ Ỹ ȧ ạ̇ ḃ ḅ ċ ċ̥ ḋ ḍ ḣ ḥ̇ k̇ k̥̇ l̇ l̥̇ ṅ ṅ̥ ṗ ṗ̥ q̇ q̥̇ ṫ ṭ̇ ṫ u̇ ẏ ỷ
Ħ̄ Ḥ̄ Ḵ̄ Ḳ̄ Ṁ Ṕ Ṗ Ṛ Ṝ Ṯ Ṯ Ṱ Ȳ Ỹ ā ạ̄ b̄ b̥̄ c̄ c̥̄ d̄ d̥̄ h̄ ḥ̄ k̄ k̥̄ l̄ l̥̄ n̄ n̥̄ p̄ p̥̄ q̄ q̥̄ t̄ ṭ̄ t̄ ū ȳ ỷ
Ḧ Ḥ̈ Ḵ̈ Ḳ̈ Ṁ Ṕ Ṗ Ṛ Ṝ Ṯ Ṯ Ṱ Ÿ Ỹ ä ạ̈ b̈ b̥̈ c̈ c̥̈ d̈ d̥̈ ẖ ḥ̈ k̈ k̥̈ l̈ l̥̈ n̈ n̥̈ p̈ p̥̈ q̈ q̥̈ ẗ ṭ̈ ẗ ü ÿ ỷ
Ħ̄ Ḥ̄ Ḵ̄ Ḳ̄ Ṁ Ṕ Ṗ Ṛ Ṝ Ṯ Ṯ Ṱ Ȳ Ỹ ā ạ̄ b̄ b̥̄ c̄ c̥̄ d̄ d̥̄ h̄ ḥ̄ k̄ k̥̄ l̄ l̥̄ n̄ n̥̄ p̄ p̥̄ q̄ q̥̄ t̄ ṭ̄ t̄ ū ȳ ỷ
Ȟ Ḥ̃ Ḵ̃ Ḳ̃ Ṁ Ṕ Ṗ Ṛ Ṝ Ṯ Ṯ Ṱ Ỹ Ỹ ã ạ̃ b̃ b̥̃ c̃ c̥̃ d̃ d̥̃ h̃ ḥ̃ k̃ k̥̃ l̃ l̥̃ ñ ñ̥ p̃ p̥̃ q̃ q̥̃ t̃ ṭ̃ t̃ ũ ỹ ỷ
Ȟ Ḥ̌ Ḵ̌ Ḳ̌ Ṁ Ṕ Ṗ Ṛ Ṝ Ṯ Ṯ Ṱ Ў Ỹ ă ặ b̆ b̥̆ c̆ c̥̆ d̆ d̥̆ h̆ ḥ̆ k̆ k̥̆ l̆ l̥̆ n̆ n̥̆ p̆ p̥̆ q̆ q̥̆ t̆ ṭ̆ t̆ ŭ ў ỷ
Ȟ Ḥ̌ Ḵ̌ Ḳ̌ Ṁ Ṕ Ṗ Ṛ Ṝ Ṯ Ṯ Ṱ Ў Ỹ ă ặ b̆ b̥̆ c̆ c̥̆ d̆ d̥̆ h̆ ḥ̆ k̆ k̥̆ l̆ l̥̆ n̆ n̥̆ p̆ p̥̆ q̆ q̥̆ t̆ ṭ̆ t̆ ŭ ў ỷ
Ȟ Ḥ̌ Ḵ̌ Ḳ̌ Ṁ Ṕ Ṗ Ṛ Ṝ Ṯ Ṯ Ṱ Ў Ỹ ă ặ b̆ b̥̆ c̆ c̥̆ d̆ d̥̆ h̆ ḥ̆ k̆ k̥̆ l̆ l̥̆ n̆ n̥̆ p̆ p̥̆ q̆ q̥̆ t̆ ṭ̆ t̆ ŭ ў ỷ
Ȟ Ḥ̌ Ḵ̌ Ḳ̌ Ṁ Ṕ Ṗ Ṛ Ṝ Ṯ Ṯ Ṱ Ў Ỹ ă ặ b̆ b̥̆ c̆ c̥̆ d̆ d̥̆ h̆ ḥ̆ k̆ k̥̆ l̆ l̥̆ n̆ n̥̆ p̆ p̥̆ q̆ q̥̆ t̆ ṭ̆ t̆ ŭ ў ỷ
Ȟ Ḥ̌ Ḵ̌ Ḳ̌ Ṁ Ṕ Ṗ Ṛ Ṝ Ṯ Ṯ Ṱ Ў Ỹ ă ặ b̆ b̥̆ c̆ c̥̆ d̆ d̥̆ h̆ ḥ̆ k̆ k̥̆ l̆ l̥̆ n̆ n̥̆ p̆ p̥̆ q̆ q̥̆ t̆ ṭ̆ t̆ ŭ ў ỷ
Ȟ Ḥ̌ Ḵ̌ Ḳ̌ Ṁ Ṕ Ṗ Ṛ Ṝ Ṯ Ṯ Ṱ Ў Ỹ ă ặ b̆ b̥̆ c̆ c̥̆ d̆ d̥̆ h̆ ḥ̆ k̆ k̥̆ l̆ l̥̆ n̆ n̥̆ p̆ p̥̆ q̆ q̥̆ t̆ ṭ̆ t̆ ŭ ў ỷ
Ȟ Ḥ̌ Ḵ̌ Ḳ̌ Ṁ Ṕ Ṗ Ṛ Ṝ Ṯ Ṯ Ṱ Ў Ỹ ă ặ b̆ b̥̆ c̆ c̥̆ d̆ d̥̆ h̆ ḥ̆ k̆ k̥̆ l̆ l̥̆ n̆ n̥̆ p̆ p̥̆ q̆ q̥̆ t̆ ṭ̆ t̆ ŭ ў ỷ

Semibold italic, Group 4, above, Above

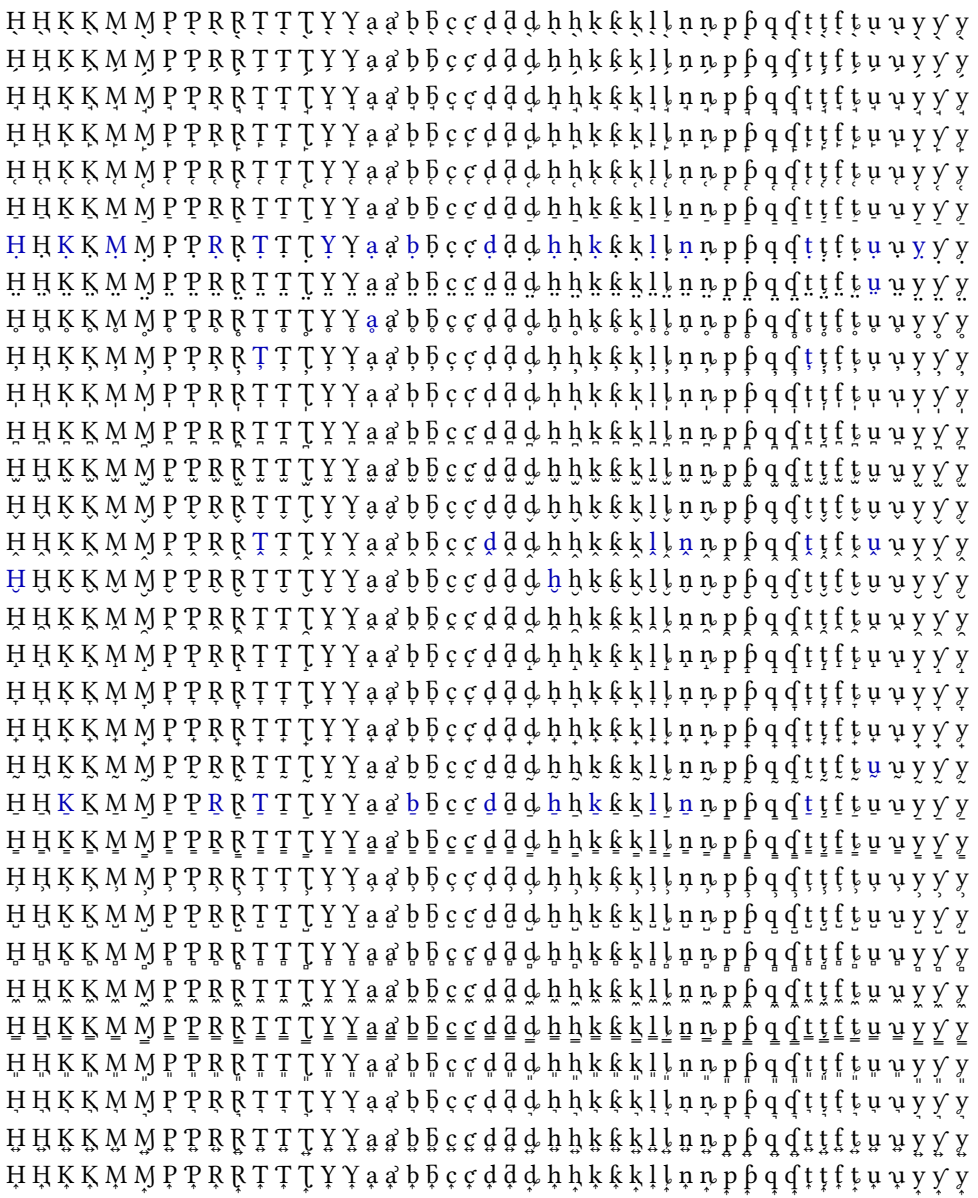


Semibold italic patched, Group 4, above, Above

Regular, Group 4, below, Below



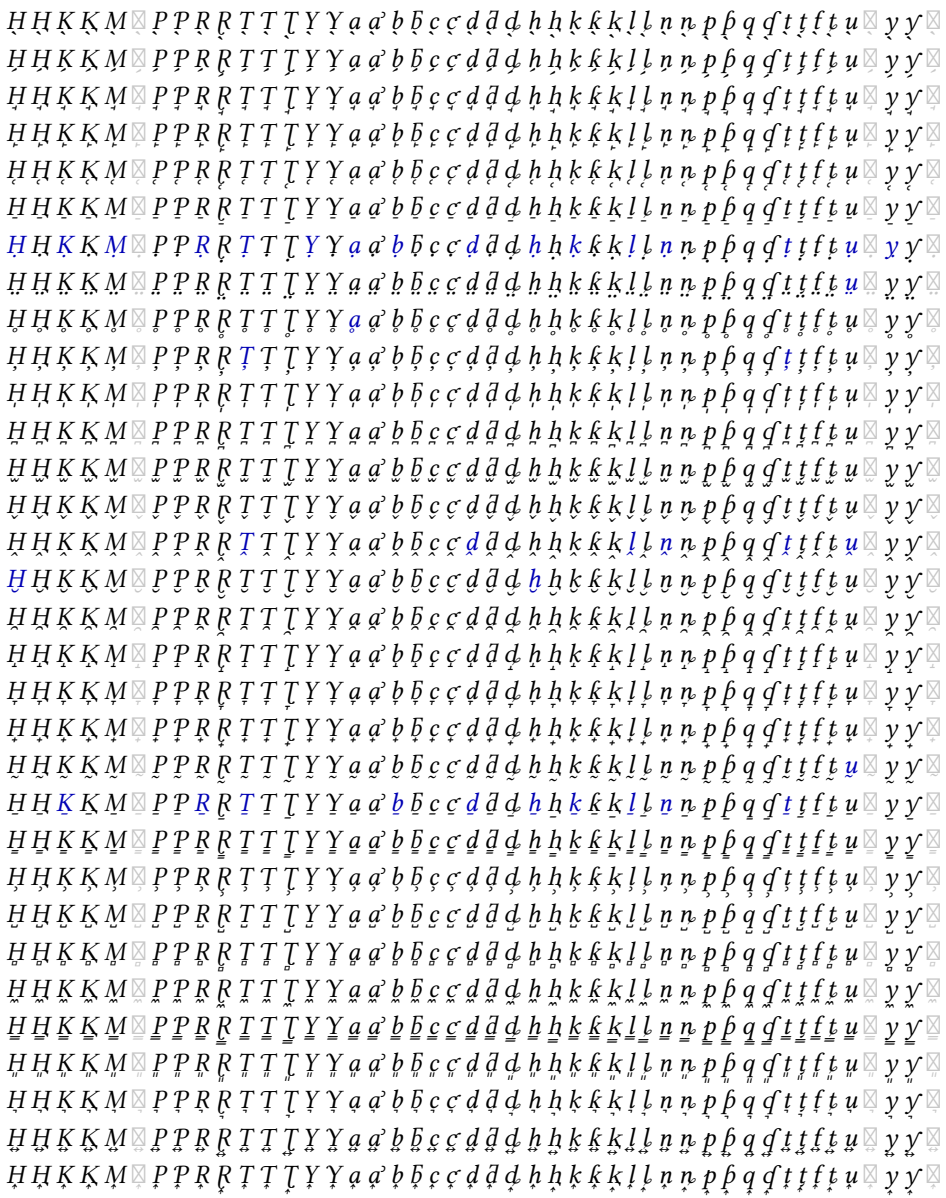
Regular patched, Group 4, below, Below



Italic, Group 4, below, Below



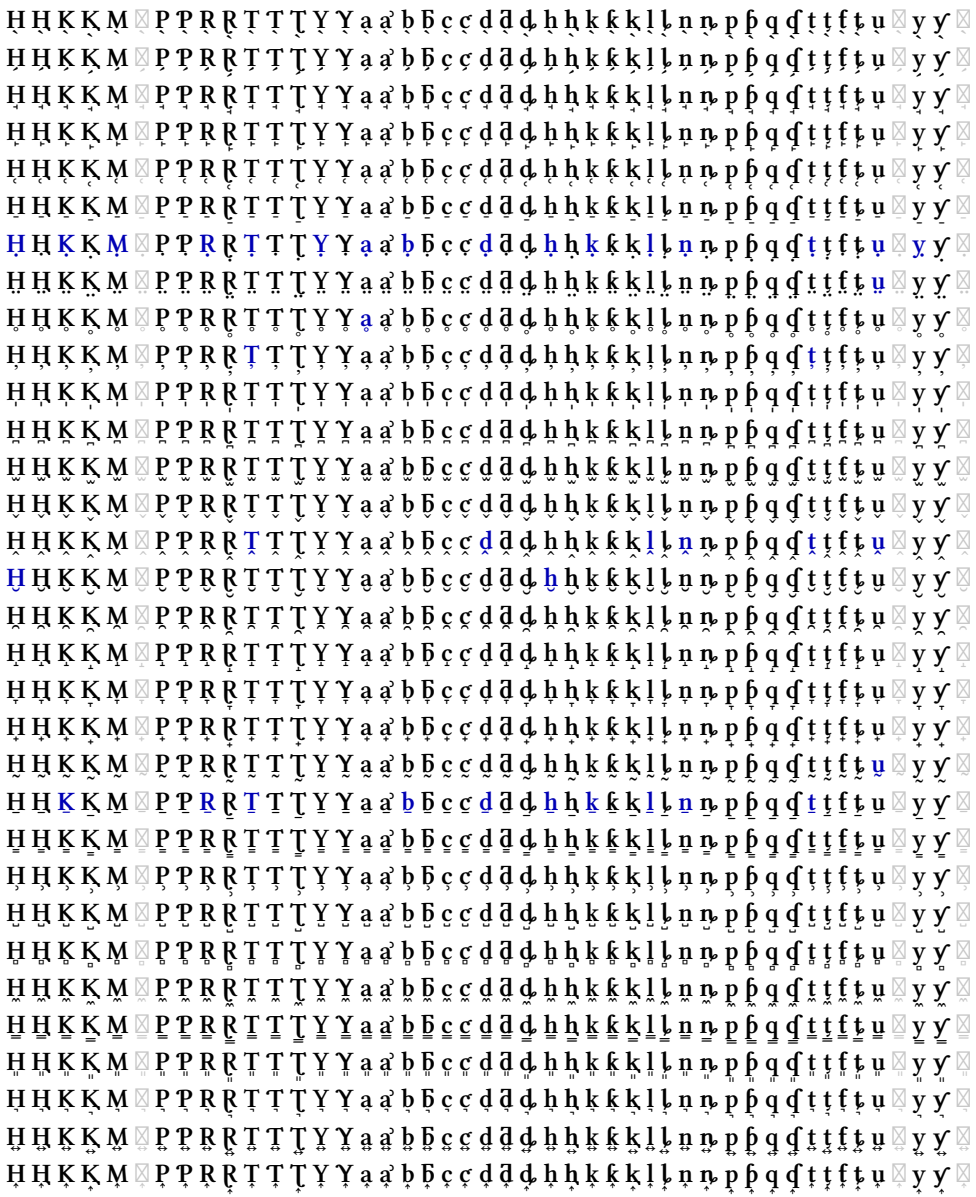
Italic patched, Group 4, below, Below



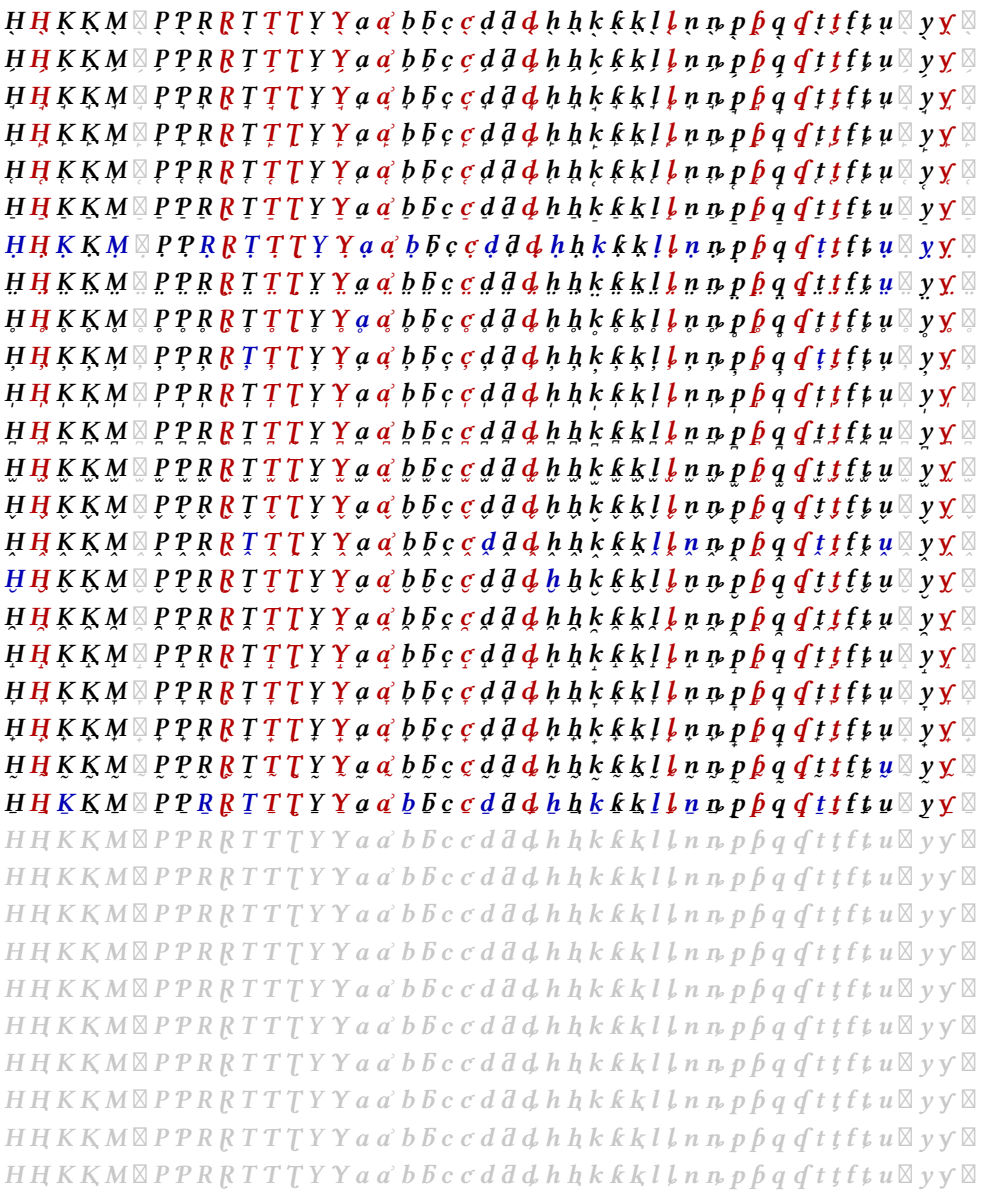
Semibold, Group 4, below, Below



Semibold patched, Group 4, below, Below



Semibold italic, Group 4, below, Below



Semibold italic patched, Group 4, below, Below

